

Modeling and Control of Flapping-Wing Micro-Aerial Vehicles With Harmonic Sinusoids

Rebecca McGill , Nak-seung Patrick Hyun , *Member, IEEE*, and Robert J. Wood, *Fellow, IEEE*

Abstract—Flapping-wing micro-aerial vehicles (FWMAVs) provide unique control challenges both in control authority and in the complexity of modeling flapping-wing aeromechanics. We provide a control-critical, closed-form model for FWMAV flight, particularly tailored to augmenting yaw torque authority of the vehicle (a common challenge for FWMAVs). The model provides invertible nonlinear closed-form expressions for thrust and moments in terms of trigonometric polynomial input voltages and the wingstroke shape, and respects asymmetries inherent to the vehicle. We apply the model to characterize the Harvard RoboBee, providing an optimization-based system identification process. Using the inverted mapping between wingstroke shape and control axes, we achieve lateral and heading control of the vehicle. Our model and asymmetric system identification, in collaboration with the chosen control scheme, allow full control authority of the position and the heading of the vehicle.

Index Terms—Aerial systems: mechanics and control, biologically-inspired robots, micro/nano robots.

I. INTRODUCTION

A PRIMARY challenge in control of flapping-wing micro-aerial vehicles (FWMAVs) lies in understanding the time-varying aerodynamic forces generated in flight. While fixed-wing vehicles can have an invertible linear mapping between input and output (e.g., quadrotors use four rotor speeds to generate thrust and three moments), the control space of FWMAVs is typically larger, including harmonic flapping trajectory shape (which may be decomposed into a Fourier series representation.)

We may compute aerodynamic forces within a wingstroke using a quasi-steady aerodynamic model, based on the blade-element method, using known kinematics and the wing shape [1]. Quasi-steady aerodynamic forces rely critically on wing pitch angles, which modulate lift and drag. However, the

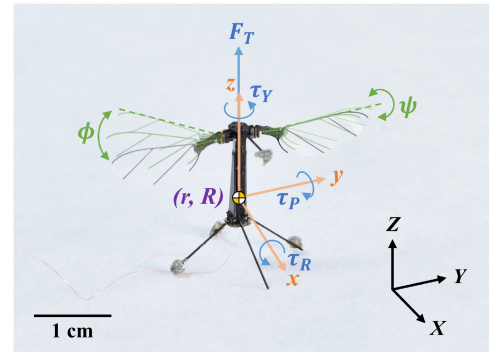


Fig. 1. The Harvard RoboBee; state described via the CoM at r with rotation matrix R (observed in flight) and wing stroke ϕ and pitch ψ angles (not directly observed), with control authority of the thrust force F_T and roll (τ_R), pitch (τ_P), and yaw (τ_Y) moments.

small scale of most FWMAVs limits observability and controllability of wing pitch (particularly for under-actuated transmission mechanisms, as in the RoboBee [2]), and vehicle weight limitations prohibit the use of encoders on the wing joint [3], [4]. For example, many FWMAVs operating at high flapping frequencies (as in [3], [5], [6]) rely on passive dynamics to achieve a desired pitch rotation. Being unobserved, passive wing pitching precludes instantaneous understanding of wing forces, so FWMAVs often use a cycle-averaged estimation of aerodynamic forces and moments for control [2], [3], [6]. (Offline, these forces may be directly measured [2], [7] or computed via the observation of wing kinematics [6].) A model with simple analytic expressions for FWMAV forces and torques based on the input wingstroke actuation would be particularly useful for control.

An additional challenge in FWMAV flight is control authority; especially for yaw torque, which is commonly the weakest control axis [8]–[10]. While two-wing FWMAVs can generate thrust, roll, and pitch with a symmetric wingstroke shape, yaw torque authority requires bilateral asymmetry to generate a non-zero cycle-averaged moment. Two-wing FWMAVs can use a “split-cycle” method to generate yaw torques, modulating the ratio of upstroke to downstroke speeds [2], [3], [11], as shown in Fig. 2. However, yaw torque authority via this method produces significantly less torque compared to the other torques (roll and pitch) [2], [7], [10], resulting in substantial obstacles for active heading control [5], [12]. Therefore, previous FWMAV controllers have required yaw-amplifying vehicle design modifications [6], [9], or have simply left heading free [5], [13].

Manuscript received July 13, 2021; accepted November 15, 2021. Date of publication December 7, 2021; date of current version December 15, 2021. This letter was recommended for publication by Associate Editor H. Hu and Editor X. Liu upon evaluation of the reviewers’ comments. This work was supported in part by the Office of Naval Research under Award #N00014-17-1-2416, in part by the National Science Foundation Graduate Research Fellowship Program under Grant DGE1745303, and under Award #IIS-1724197, and in part by the Wyss Institute for Biologically Inspired Engineering, with prototypes enabled by equipment supported by the ARO DURIP program under Award #W911NF-13-1-0311. (Corresponding author: Rebecca McGill.)

The authors are with the John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, MA 02138 USA (e-mail: rhsteinmeyer@g.harvard.edu; nphyun@seas.harvard.edu; rjwood@eecs.harvard.edu).

This letter has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2021.3132921>, provided by the authors.

This letter has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2021.3132921>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2021.3132921

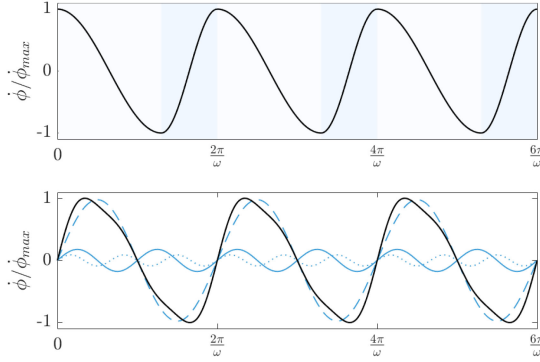


Fig. 2. Flapping trajectories (wing velocity $\dot{\phi}$) used to generate yaw torque for FWMAVs; the split-cycle method (top) and second harmonic injection method (bottom), with dashed fundamental frequency, solid second harmonic, and dotted third harmonic.

Alternately, yaw may be generated by varying the second harmonic of the flapping signal, combined with a sub-resonant fundamental frequency (to reduce passive filtering of higher-order harmonics) [14]. This second harmonic injection method achieves a similar wingstroke shape to the split-cycle method, without piece-wise sinusoid concatenation (see Fig. 2). While this method has the potential to dramatically improve FWMAV control authority, it begets challenges in modeling. While a closed-form model for drag force (and by extension, yaw torque) exists for split-cycle flapping [2], deriving a closed-form, analytic model for the second harmonic injection method effectively requires finding the roots of a high-degree (≥ 6) trigonometric polynomial, which is impossible per the Abel-Ruffini theorem [15]. Recent work has verified the second harmonic injection method for yaw torque in flight [6], but its corresponding lift and drag forces were found numerically, using data from video of the flapping of a constrained (static) vehicle.

A further challenge in FWMAV control is the presence of asymmetries introduced into the vehicle during manufacturing [7], which result in sizeable control offsets in flight. Whereas FWMAV models do not commonly account for imperfections inherent to a vehicle, assuming symmetry despite the likelihood of manufacturing anomalies [2], [16], control design strategies attempt to mitigate dynamic effects resulting from asymmetries via force and torque offsets within the controller, whether constant or adaptive [7], [17]. However, these strategies serve to mask the characteristics of a vehicle, rather than to understand or even exploit them in control.

In this paper, we derive an analytic closed-form, invertible model describing cycle-average thrust and torques for a FWMAV driven by multiple sinusoidal control signals, allowing full vehicle control. Our method allows yaw control via the second harmonic injection method, and we include the effects of this method on thrust and the three body torques. We also include asymmetries due to physical discrepancies between opposite actuator-wing pairs. We provide an optimization-based system identification method to characterize a FWMAV, including its asymmetries, using the nonlinear model, applying open-loop

data from 40 unique flights. Our mapping allows general controllers (e.g., those designed for quadrotors) to directly control a FWMAV. We apply a geometric, coordinate-free controller from [18] to the Harvard RoboBee; a two-wing, 86 mg FWMAV. We apply active heading control both while hovering in place and while tracking a trajectory, to demonstrate in-flight heading control for the RoboBee, and achieve full control authority for the first time.

II. FWMAV MODELING WITH CYCLE-AVERAGED AERODYNAMICS

The propulsion of FWMAVs relies on aerodynamic lift and drag forces generated during flapping. The blade element based quasi-steady aerodynamic model is commonly used to compute the instantaneous quasi-steady aerodynamic force within a cycle, though it requires knowledge of wing configurations (the wing stroke angle ϕ , and the wing pitch angle ψ), which is not available during controlled flight [5]–[7]. Therefore, FWMAV models commonly rely on simplifying assumptions to identify a mapping between actuator inputs (i.e., harmonic voltage signals) and the harmonic wing trajectory.

Assumption 1: The body dynamics and the wing dynamics are decoupled.

Assumption 2: Body dynamics depend on cycle-averaged (time-independent) aerodynamic forces.

Assumption 1 applies to FWMAVs whose wing mass is markedly lower than the body mass (e.g., the wing mass of the Harvard RoboBee contributes to less than 1% of its total mass). Further analysis of the contribution of wing inertia to whole body dynamics may be found in [19] and [20]. Assumption 2 applies to FWMAVs characterized by high-frequency flapping, relative to slower body dynamics. (The Harvard RoboBee typically flaps at 160 Hz, the RoboFly at 170 Hz, [6], and Bee+ at 100 Hz [8]). This assumption allows identification of time-independent estimations for lift and drag forces, for practical use in controlled flight [3], [4], [6], [8].

A. Cycle-Averaged Quasi-Steady Aerodynamic Forces

Consider harmonic wing trajectories, with stroke angle $\phi_i(t)$ and a harmonic wing pitch angle $\psi_i(t)$ given for each i -th wing, where $i \in \{L, R\}$ indicates the left and right wings, respectively. Let T be the period of the fundamental of the harmonic wing stroke motion, $\phi(t)$, and $\omega := 2\pi/T$ is the flapping frequency. Then, the blade-element method can be used to compute the cycle-averaged lift and drag force ($\bar{F}_L^i$ and $\bar{F}_D^i$) produced by harmonic wing flapping [1], [2],

$$\bar{F}_L^i := \frac{\omega\rho\beta}{4\pi} \int_0^{2\pi} C_L(\alpha_i) \dot{\phi}_i^2 dt \quad (1)$$

$$\bar{F}_D^i := \frac{\omega\rho\beta}{4\pi} \int_0^{2\pi} C_D(\alpha_i) \dot{\phi}_i |\dot{\phi}_i| dt. \quad (2)$$

with angle of attack $\alpha_i(t)$ (varying with wing pitch ψ). The wing stroke velocity is $\dot{\phi}_i$, the ambient air density is given by ρ , and the wing geometry parameter is β . We use experimentally-identified

lift and drag coefficients, $C_L(\alpha_i)$ and $C_D(\alpha_i)$, from a *Drosophila* wing [1], [21].

B. Cycle-Averaged FWMAV Dynamics

Suppose that $r = (x, y, z)$ is the center of mass (CoM) of the vehicle, and the origin of the body-attached coordinate frame is located at the CoM as shown in Fig. 1. We align the body frame with the principal axis of the vehicle, and model the vehicle body as a rigid body with (r, R) (for rotation matrix R) as a homogeneous coordinate in $SE(3)$, the Special Euclidean Group, which is a semi-direct product between $\mathbb{R}^3$ and $SO(3)$, the Special Orthogonal Group.

$$\dot{r} = v \quad (3)$$

$$\dot{R} = R\hat{\Omega} \quad (4)$$

$$m\dot{v} = -mge_3 + Re_3F_T \quad (5)$$

$$I\dot{\Omega} = -\Omega \times I\Omega + \tau, \quad (6)$$

The vector $e_3 = (0, 0, 1)^\top$ is a unit vector in $\mathbb{R}^3$, m is the vehicle mass, g is gravitational acceleration, F_T is the thrust force, and $\tau = (\tau_r, \tau_p, \tau_y)$ is a vector of torques along the roll, pitch, and yaw axes. The principal moment of inertia, $I \in \mathbb{R}^{3 \times 3}$, is a diagonal matrix with components (I_{xx}, I_{yy}, I_{zz}) , and $\hat{\Omega}$ is a skew-symmetric matrix such that, $\hat{\Omega}y = \Omega \times y$ (here, $\times$ indicates a cross product). For the RoboBee, we use $m = 86$ mg (measured), $I_{xx} = 1.42$ g·mm², $I_{yy} = 1.34$ g·mm², and $I_{zz} = 0.45$ g·mm² [5]. We assume that the direction of the thrust F_T is aligned with the body z axis.

The thrust and the torques are generated by the aerodynamic forces acting on each wing. Following Assumption 2, we may estimate the generated thrust and torques using the cycle-averaged lift and drag forces:

$$F_T = \bar{F}_L^L + \bar{F}_L^R \quad (7)$$

$$\tau_r = r_{cp}(\cos(\phi_{mean}^L)\bar{F}_L^L - \cos(\phi_{mean}^R)\bar{F}_L^R) \quad (8)$$

$$\tau_p = r_{cp}(\bar{F}_L^L \sin(\phi_{mean}^L) + \bar{F}_L^R \sin(\phi_{mean}^R)) \quad (9)$$

$$\tau_y = r_{cp}(\bar{F}_D^L - \bar{F}_D^R). \quad (10)$$

Distance from the body z -axis to the wing center of pressure is r_{cp} , and average stroke angle is $\phi_{mean}^{L,R}$.

III. ANALYTIC MAPPING FROM INPUT VOLTAGE TO THRUST AND TORQUE WITH MULTIPLE SINUSOIDS

In designing our model for FWMAVs, we consider sinusoidal flapping signals composed with three harmonics and one DC component, per Eq. (12). Also, we give special consideration to inherent vehicle asymmetries often introduced during manufacturing, which are nearly impossible to avoid.

A. Input Voltage to Wing Trajectory Approximation

We apply the second-order linear system approximation describing actuator and transmission dynamics in [22], which models the transfer function (from the harmonic input voltage

signal to the harmonic trajectories of the wing stroke) using an equivalent mass, m_{eq} , damping coefficient, b_{eq} , and stiffness k_{eq} . This transfer function H characterizes the actuator and transmission dynamics at a flapping frequency ω as follows:

$$H(\omega) = \frac{A_H}{m_{eq}(\omega j)^2 + b_{eq}(\omega j) + k_{eq}}, \quad (11)$$

where $\Phi(\omega) = H(\omega)V(\omega)$ for Φ and V , representing the Fourier transform of the wing stroke angle ϕ and the input voltage signal $\mathcal{V}$, respectively. We actuate each wing by applying a harmonic voltage signal

$$\mathcal{V}^i(t) = V_{off}^i + V_{amp}^i \sum_{n=1}^3 a_n \sin(n\omega t) \quad (12)$$

for each $i \in \{L, R\}$, where a_n are Fourier coefficients which determine the shape of the harmonic signals, V_{amp} is the signal amplitude, and V_{off} is the DC offset. The voltage signals to each wing are composed of superimposed sinusoids at the fundamental flapping frequency and its second and third harmonics, as shown in Fig. 2, with corresponding coefficients $a_i, i \in [1, 3]$. We select each a_i such that a_1 primarily influences wingstroke amplitude, a_2 biases the relative speeds of the upstroke to downstroke and induces a net yaw torque, and a_3 flattens and smooths the sinusoid, decreasing voltage requirements for a given a_1 [23]. In practice, we choose constant a_1 and a_3 , with signal shape controlled by a_2 . We choose a_1 and a_3 such that the maximum amplitude of the base shape function $\sum_{n=1}^3 a_n \sin(n\omega t)$ is one, for some nominal $a_2 = 0$ (zero yaw condition).

The instantaneous wingstroke angle, ϕ , at time t is

$$\begin{aligned} \phi(t) = & (V_{off} + \nu)|H(0)|\text{sgn}\angle H(0) \\ & + V_{amp} \sum_{n=1}^3 a_n |H(n\omega)| \sin(n\omega t - \angle H(n\omega)), \end{aligned} \quad (13)$$

where $|\cdot|$ is the absolute value. In practice, the mean-stroke angle, ϕ_{mean} (DC component of (13)), has a bias introduced during actuator manufacturing, where ν is the required offset voltage to compensate this bias.

B. Closed-Form Cycle-Averaged Lift Force

We compute the cycle-averaged aerodynamic lift and drag forces in (1)–(2), with the second-order expression for wing stroke angle ϕ^i from (13), as a function of tunable parameters such as the amplitude V_{amp} , the offset V_{off} , and harmonic shape parameters $\{a_n\}_{n=1}^3$. However, acquiring closed-form expressions is a substantial challenge, in part due to the lack of observability of the wing pitch. To proceed in deriving a mapping from the input voltage to the cycle-averaged thrust and torques, we use the following assumption:

Assumption 3 ([2]): We use a constant angle-of-attack (AoA) in the cycle-averaged aerodynamic force and drag calculations; $\dot{\alpha}^i = 0$.

Using this assumption, we find estimated lift and drag forces (which hold for any constant AoA), enabling us to provide analytical closed-form expressions for time-averaged thrust and torques. We learn the lift and drag gains through the

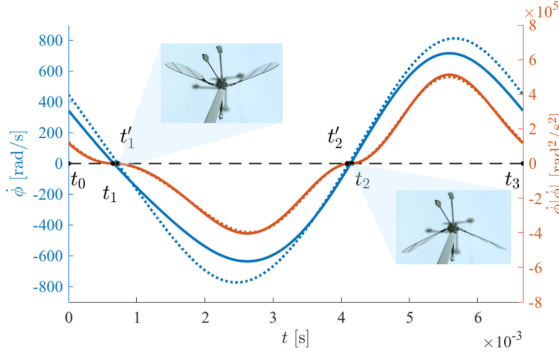


Fig. 3. Visualization of model assumptions: (blue) wingstroke speed $\dot{\phi}$, with full $\dot{\phi}$ shown with a solid line, and fundamental frequency approximation shown with a dotted line; (red) full $\dot{\phi}|\dot{\phi}|$ shown using a solid line, and simplification (with third harmonic term removed) shown with a dotted line. (For $\dot{\phi}$ in this figure, we use $\omega = 150$ Hz, $V_{amp} = 180$ V, $a_1 = 1.1$, $a_2 = 0.1$, and $a_3 = 0.1$.)

optimization-based system identification process in Section IV. Solving (1) for a closed-form $\bar{F}_L$, applying Assumption 3 with wingstroke angle $\phi(t)$ per (13), yields

$$F_L = \frac{1}{4} \rho \beta C_L \omega^2 V_{amp}^2 \sum_{n=1}^3 n^2 (a_n |H(n\omega)|)^2, \quad (14)$$

allowing analytic expressions for thrust, roll torque, and pitch torque. However, finding a closed-form solution for the drag force (and by extension, yaw torque) remains a challenge even with a constant AoA (per Assumption 3) due to the sign changes in the integrand of (2).

C. Closed-Form Cycle-Averaged Drag Force

While the second-harmonic drag and yaw torque method provides substantial modeling challenges, we maintain that it provides distinct advantages over the split-cycle method in that it is fully analytic and allows instantaneous, continuous yaw commands.

We first find an appropriate approximation for the closed-form drag force, $\bar{F}_D$, according to the integral in (2) and using Assumption 3. A representative example of $\dot{\phi}|\dot{\phi}|$ of the RoboBee flap in the fixed platform is illustrated in Fig. 3. During each wingstroke cycle, there always exist two zero crossings within the interval $[t_0, t_0 + 2\pi/\omega)$. Without loss of generality, we assign $t_0 = 0$, $t_3 = \frac{2\pi}{\omega}$, and zero-crossing points of $\dot{\phi}|\dot{\phi}|$ to be t_1 and t_2 (see Fig. 3). Suppose $\dot{\phi}(t) \geq 0$ for all $t \in [t_0, t_1] \cup [t_2, t_3]$ and $\dot{\phi}(t) \leq 0$ for all $t \in [t_1, t_2]$. Let Q be the anti-derivative of $\dot{\phi}^2$, which can be explicitly computed using (13) ($Q := \int \dot{\phi}^2$), then

$$\int_0^{\frac{2\pi}{\omega}} \dot{\phi}|\dot{\phi}| dt = Q(t_3) - Q(t_0) + 2(Q(t_1) - Q(t_2)). \quad (15)$$

Thus, the closed form for $\bar{F}_D$ depends on explicit solutions at zero-crossings t_1 and t_2 of $\dot{\phi}(t_i) = 0$. However, finding direct solutions for t_1 and t_2 given $\dot{\phi}$ requires a closed-form solution

for this six-degree trigonometric polynomial: Given its arbitrary phases, this is not possible (per the Abel-Ruffini theorem [15]).

Therefore, we approximate t_1 and t_2 , using zero-crossings of the fundamental, $V_{amp}[A \cos(\omega t - a)]$. This yields approximate roots of $\dot{\phi}(t) = 0$, with $t'_1 = \frac{2a+\pi}{2\omega}$ and $t'_2 = \frac{2a+3\pi}{2\omega}$. Then, we evaluate the antiderivative, $Q(t)$ at each t'_i . Applying these within (15) provides an analytical approximation for the cycle-averaged drag force, per (2):

$$\begin{aligned} \bar{F}_D \approx \frac{\rho \beta V_{amp}^2 C_D}{60\pi} & [40AB \cos(2a - b) \\ & - AC \sin(3a - c) \\ & + 12BC (\cos(5a - b - c) \\ & + 5 \cos(a + b - c))] , \end{aligned} \quad (16)$$

where $A = a_1 \omega |H(\omega)|$, $a = \angle H(\omega)$, $B = 2a_2 \omega |H(2\omega)|$, $b = \angle H(2\omega)$, $C = 3a_3 \omega |H(3\omega)|$, and $c = \angle H(3\omega)$, allowing a closed-form expression for yaw torque per (10).

D. Full Closed-Form Mapping of Input Voltage to Force/Torques Mapping, Including Yaw Torque

Before providing a full closed-form expression for drag force, we make one final simplification: the magnitude of the transfer function, $H(\omega)$ in (11), for the RoboBee at the third harmonic, $|H(3\omega)|$, is significantly smaller than that of the lower harmonics [2], [14]. The third harmonic term has only a very small contribution to lift and drag; instead, it serves to flatten the input voltage signal [14]. (Coefficient a_3 is very small compared to a_1 ; 10% in [14].) It follows that the contribution to the cycle-averaged drag in (16) from $C = 3a_3 \omega |H(3\omega)|$ is relatively small, which allows us to assume $C = 0$ in (16). Finally, since the mean stroke angle ϕ_{mean} (in (8)–(9)) is very small (less than ± 10 deg), we may apply small angle approximations $\sin(\phi_{mean}) \approx \phi_{mean}$ and $\cos(\phi_{mean}) \approx 1$ in the final expressions as in [2].

Suppose that the shape parameters for the left and right input voltage signals in (12), $\{a_i^L\}_{i=1}^3$ and $\{a_i^R\}_{i=1}^3$, are symmetrical, such that $a_1^L = a_1^R$, $a_2^L = -a_2^R$, and $a_3^L = a_3^R$. Then, we may write closed-form expressions for the thrust, F_T , and three moments, (τ_r, τ_p, τ_y) as a function of five input parameters: two amplitudes of the first order harmonic, $V_L := V_{amp}^L$ and $V_R := V_{amp}^R$, two offset voltages, V_{off}^L and V_{off}^R , and the shape parameter for the second order harmonic, a_2 . Defining $\zeta = a_1^2 |H(\omega)|^2 + 4a_2^2 |H(2\omega)|^2$, we have:

$$F_T \approx \frac{1}{4} \rho \beta C_L \omega^2 (\eta V_L^2 + V_R^2) \zeta \quad (17)$$

$$\tau_r \approx r_{cp} \rho \beta C_L \omega^2 (\eta V_L^2 - V_R^2) \zeta \quad (18)$$

$$\tau_p \approx r_{cp} \rho \beta C_L |H(0)| \text{sgn} \angle H(0) \omega^2 \quad (19)$$

$$\times (\eta V_L^2 (V_{off}^L + \nu) + V_R^2 (V_{off}^R + \nu)) \zeta$$

$$\begin{aligned} \tau_y \approx \frac{4}{3\pi} r_{cp} \rho \beta C_D \omega^2 a_1 (a_2 + \mu) |H(\omega)| |H(2\omega)| \\ \times \cos(2\angle H(\omega) - \angle H(2\omega)) (\eta V_L^2 + V_R^2). \end{aligned} \quad (20)$$

We also introduce a yaw offset term, μ (similar to ν , which accounts for mechanical bias in the pitch offsets), to describe the effective change in a_2 due to wing asymmetries and imperfections introduced during manufacturing. In addition, we include the parameter η to account for discrepancies in force production between wing-actuator pairs on the left and right sides. (It is practically infeasible, when manufacturing, for the two sides to be identical.) Asymmetry parameters η , ν , and μ are found via the following optimization process.

IV. OPTIMIZATION-BASED SYSTEM IDENTIFICATION

To characterize a given FWMAV according to the model described in Section III, we apply an optimization-based system identification process using open-loop data. The closed-form mapping from the input voltage to the thrust and moments in (17)–(20) requires knowledge of physical parameters including air density, ρ , wing geometry β , lift and drag coefficients, C_L and C_D , and the transfer function $H(\omega)$. We use a motion capture system (Vicon T040 System, Oxford U.K.) to collect open-loop flight data to numerically estimate cycle-averaged thrust and moments.

In anticipation of control applications, and to simplify the model, we first rephrase vehicle forces and torques in terms of direct wing-flapping signal inputs and grouped physical parameters. We choose control inputs u_i , $i \in [1, 5]$, as follows: $u_1 = V_L^2$, $u_2 = V_R^2$, $u_3 = a_2$, $u_4 = V_{off}^L$, and $u_5 = V_{off}^R$. These parameters allow independent modification of thrust and body torques. Next, we choose grouped parameters γ_i , $i \in [1, 3]$ and δ_i , $i \in [1, 3]$ (grouped such that δ_i parameters are easier to directly measure, for use in system identification and control). We assign $\gamma_1 = \frac{1}{4}\rho\beta C_L\omega^2$, $\gamma_2 = r_{cp}$, and $\gamma_3 = \frac{C_D}{C_L} \cdot \frac{8}{3\pi} \cos(2\angle H(\omega) - \angle H(2\omega))$; and $\delta_1 = a_1\omega|H(\omega)|$, $\delta_2 = 2\omega|H(2\omega)|$, and $\delta_3 = |H(0)|\text{sgn}\angle H(0)^*$. With the constant lift and drag coefficients assumption, along with Assumption 3, our model becomes

$$F_T = \gamma_1(\delta_1^2 + \delta_2^2 u_3^2)(\eta u_1 + u_2) \quad (21)$$

$$\tau_r = \gamma_1\gamma_2(\delta_1^2 + \delta_2^2 u_3^2)(\eta u_1 - u_2) \quad (22)$$

$$\tau_p = \gamma_1\gamma_2\delta_3(\delta_1^2 + \delta_2^2 u_3^2) \quad (23)$$

$$\times (\eta u_1(u_4 + \nu) + u_2(u_5 + \nu))$$

$$\tau_y = \gamma_1\gamma_2\gamma_3\delta_1\delta_2(u_3 + \mu)(\eta u_1 + u_2). \quad (24)$$

In preparation for the optimization-based system identification process that follows, we separate the parameters from (21)–(24) into four groups, as follows:

$$\bar{p}_1 := (\gamma_1, \gamma_2, \eta), \bar{p}_2 := (\nu, \delta_3), \bar{p}_3 := (\mu, \gamma_3) \quad (25)$$

$$\bar{q} := (\delta_1, \delta_2) \quad (26)$$

We assume $\{\bar{p}_i\}_{i=1}^3$ are unknown parameters, related to the lift and drag coefficients and vehicle asymmetry, and $\bar{q}$ are known parameters, found using nominal actuator and transmission physical parameters (provided in [14]).

A. Cycle-Averaged Thrust and Moment Measurement

Applying the closed-form model to characterize the RoboBee, we measure vehicle forces and torques from dynamic tests with a variety of wingstroke control conditions, using this data to find an optimal fitting between our model and the data. Since the cycle-averaged dynamics described in (3)–(6) are markedly slower than the flapping frequency (150 Hz or higher), and the Vicon motion-capture system, at 500 Hz, is fast enough to sample the relatively slow-varying body dynamics, we can estimate the cycle-averaged thrust and moments using the motion-capture data recorded during flight.

First, we gathered a comprehensive set of 40 open-loop flights (0.35 seconds), varying control parameters to achieve measurable thrust and body torque combinations, with a 150 Hz flapping frequency (approximately 15 Hz below resonance for our vehicle). Flights (open-loop, and later controlled) were contained within a motion-capture arena with position and attitude captured using the Vicon system, and power and control lines supplied from off-board computers and amplifiers.

We calculate acceleration as the vehicle lifts off (the first 25 ms of flight; approximately four cycles of flapping data). We find thrust and body torques using the low-pass filtered derivatives,

$$F_T^e = \langle \bar{R}e_3, m\dot{\bar{v}} + mge_3 \rangle \quad (27)$$

$$\tau^e = \dot{I}\bar{\Omega} + I\bar{\Omega} \times \bar{\Omega} \quad (28)$$

where $\langle \cdot, \cdot \rangle$ is the inner-product in $\mathbb{R}^2$; and $\bar{R}$, $\bar{v}$, and $\bar{\Omega}$ are Vicon data (cycle-averaged, with filtered derivatives).

B. Optimization-Based System Identification

We design an optimization-based mapping process between FWMAV data and the model in (21)–(24), which is characterized by nonlinear relationships between unknown parameters $\{\bar{p}_i\}_{i=1}^3$ (grouped in (25)). We observe that the closed-form model has a nearly triangular structure, as the number of unknown parameters in the closed form increases from F_T to τ_y . Therefore, we formulate an optimization sequence, identifying the optimal parameters for $\bar{p}_1^* := (\gamma_1^*, \gamma_2^*, \eta^*)$ first, then using the optimal $\bar{p}_1^*$ to find optimal parameters for $\bar{p}_2$ and $\bar{p}_3$. To find $\bar{p}_1^*$, we formulate a nonlinear optimization problem, minimizing the weighted sum of estimation errors for F_T and τ_r . Suppose that $\chi = \gamma_1 * \eta$ and $\bar{F}_T^e, \bar{\tau}_r^e \in \mathbb{R}^N$ are vectorized measured thrust and roll torque from N open-loop flights. Then, our problem is

$$(\gamma_1^*, \chi^*, \eta) := \arg \min_{\bar{p}_1 \in \mathbb{R}^3} J(\bar{p}_1)$$

with cost function

$$J(\bar{p}_1) := \|\bar{F}_T^e - \bar{F}_T(\bar{p}_1)\|_2^2 + \lambda \|\bar{\tau}_r^e - \bar{\tau}_r(\bar{p}_1)\|_2^2,$$

using tunable weight $\lambda > 0$, where $\|\cdot\|_2$ is the 2-norm. $\bar{F}_T$ and $\bar{\tau}_r$ are vectorized estimates (according to the closed-form model) in $\mathbb{R}^N$ using the commanded control inputs $\{u_i\}_{i=1}^5$.

The change of variables using χ allows us to reformulate the expression for F_T to have a linear dependence on γ_1 and χ ,

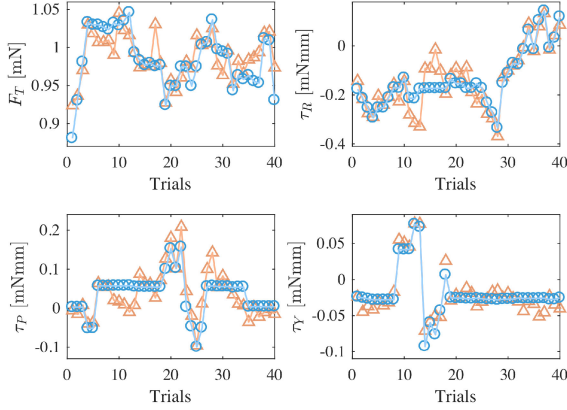


Fig. 4. Measured (orange) and model-estimated (blue) forces and torques over 40 open-loop flights, using optimal system identification.

maintaining bilinearity only in $\bar{\tau}_r$:

$$\bar{F}_T(\bar{p}_1) = \Delta_1 \begin{pmatrix} \gamma_1 \\ \chi \end{pmatrix}, \bar{\tau}_r(\bar{p}_1) = \gamma_2 \Delta_2 \begin{pmatrix} \gamma_1 \\ \chi \end{pmatrix}$$

for known $\Delta_1, \Delta_2 \in \mathbb{R}^{N \times 2}$. Let $\bar{p}_{11}^* = (\gamma_1^*, \chi^*)$ be the partial state of the optimal solution $\bar{p}_1^*$. Then, we have a nominal solution of $\gamma_2^* = (\Delta_2 \bar{p}_{11}^*)^\dagger \bar{\tau}_r^e$ and

$$((\bar{F}_T^e)^\top \Delta_1 + \lambda (\bar{\tau}_r^e)^\top \Delta_2 \gamma_2^*) = \bar{p}_{11}^{*\top} (\Delta_1 \Delta_1^\top + \lambda \Delta_2 \Delta_2^\top \gamma_2^{*2})$$

where $\dagger$ is the pseudo-inverse. By numerically solving the above necessary condition, we obtain the optimal $\bar{p}_1^*$.

Next, we solve two independent optimization problems to find the optimal $\bar{p}_2^* = (\nu^*, \delta_3)$ and $\bar{p}_3^* = (\mu, \gamma_3)$:

$$(\nu^*, \delta_3^*) := \arg \min_{\bar{p}_2 \in \mathbb{R}^2} \|\bar{\tau}_p^e - \bar{\tau}_p(\bar{p}_2, \bar{p}_1^*)\|_2^2$$

$$(\mu^*, \gamma_3^*) := \arg \min_{\bar{p}_3 \in \mathbb{R}^2} \|\bar{\tau}_y^e - \bar{\tau}_y(\bar{p}_3, \bar{p}_1^*)\|_2^2.$$

We provide the model fitting results in Fig. 4. The model predicts the trends in thrust and torques with varied control inputs, and we find that the root-mean-square error across all trials between the model-predicted and measured values are 0.0220 mN for thrust force (18.1% of the range of measured values), 0.0540 mNmm for roll torque (10.8%), 0.0328 mNmm for pitch torque (10.8%), and 0.0116 mNmm for yaw torque (7.8%).

V. GEOMETRIC CONTROL WITH INVERSE MAPPING

With this closed-form model for FWMAV flight, fit to our vehicle in Section IV, we perform a lateral and heading control task. Previous RoboBee controllers in [5], [12], [13] only tracked reference positions, with a free heading angle (simply minimizing yaw velocity).

A. Inverse Mapping for Feedback Control

We produce an analytical inverse of the model, to allow direct calculation of control parameters required for desired forces and torques (respecting vehicle asymmetries, and thus not requiring unmodeled offsets). We have five control parameters, $\{u_i\}_{i=1}^5$

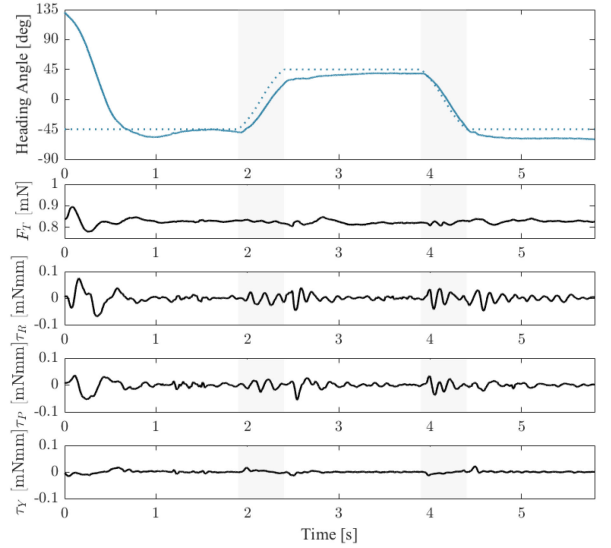


Fig. 5. Heading response for tracking a desired heading angle during hovering flight, with estimated forces and torques (from motion-capture data) during the control response.

to yield four desired outputs (one thrust and three moments) per (21)–(24). This indicates that the mapping is surjective, but not injective. Thus, there exists an extra degree of freedom to generate the same desired output. We simplify our inverse mapping by choosing symmetrical voltage offsets, $V_{off}^L = V_{off}^R$ (i.e., $u_4 = u_5$). For simplicity, we neglect the contributions of μ to u_1 and u_2 . We compute the following nonlinear mapping using the MATLAB symbolic solver to directly find control inputs for desired thrust and torque set $(F_T, \tau_r, \tau_p, \tau_y) \in \mathbb{R}^+ \times \mathbb{R}^3$:

$$u_1 = \frac{(\gamma_2 \gamma_3 F_T + (\gamma_2^2 \gamma_3^2 F_T^2 - 4 \tau_y^2)^{1/2}) (\gamma_2 F_T + \tau_r)}{4 \eta \gamma_1 \gamma_2^2 \gamma_3 \delta_1^2 F_T} \quad (29)$$

$$u_2 = \frac{(\gamma_2 \gamma_3 F_T + (\gamma_2^2 \gamma_3^2 F_T^2 - 4 \tau_y^2)^{1/2}) (\gamma_2 F_T - \tau_r)}{4 \gamma_1 \gamma_2^2 \gamma_3 \delta_1^2 F_T} \quad (30)$$

$$u_3 = \frac{\delta_1 (\gamma_2 \gamma_3 F_T - (\gamma_2^2 \gamma_3^2 F_T^2 - 4 \tau_y^2)^{1/2})}{2 \delta_2 \tau_y} - \mu \quad (31)$$

$$u_4 = \frac{\tau_p}{\gamma_2 \delta_3 F_T} - \nu, \quad (32)$$

with a valid domain for $F_T > 0$. If $\tau_y = 0$, applying L'Hôpital's rule for the limit as $\tau_y \rightarrow 0$ yields $u_3 = -\mu$.

B. Geometric Position and Heading Control

With the control authority observed in Section IV, we anticipate that our vehicle is equipped with the ability to operate according to quadrotor-like control methods on $SE(3)$. We apply a geometric tracking controller on $SE(3)$ following [18], which provides coordinate-free asymptotic control of CoM position and heading. Suppose that $r_d(t)$ is the desired CoM trajectory, and $b_1^d(t) \in \mathbb{R}^3$ is the desired heading orientation in the inertial frame (defined by the x -axis in Fig. 1) at time t . Then, the

feedback control law is defined as

$$\begin{aligned} F_T &= \langle -k_x e_x - k_v e_v + m g e_3 + m \ddot{r}_d, R e_3 \rangle \\ \tau &= -k_R e_R - k_\Omega e_\Omega + \Omega \times I \Omega \\ &\quad - I(\dot{\Omega} R^\top R_d \Omega_d - R^\top R_d \dot{\Omega}_d), \end{aligned}$$

where $e_x := r - r_d$, $e_v := v - \dot{r}_d$, R_d is the desired orientation, and Ω_d is the desired angular velocity. R_d and Ω_d are uniquely defined with the desired thrust vector and the desired heading b_1^d (see [18] for more details). Orientation error is captured by $e_R := 1/2(R_d^\top R - R^\top R_d)^\vee$ (where $\vee$ is the inverse of the skew-symmetric hat map of a matrix), which represents the distance metric on $SO(3)$; Angular velocity error is captured by $e_\Omega := \Omega - R^\top R_d \Omega_d$, related to the derivative of $R_d^\top R$; and $k_x, k_v, k_R, K_\Omega \in \mathbb{R}^{3 \times 3}$ are diagonal matrices with positive entries representing controller gains.

C. Hovering Flight With Variable Heading Angle

We apply this controller during a hovering flight, commanding a fixed position, $r_d = (-0.012, -0.034, 0.14)$ mm, and tracking a variable heading vector. We command heading vectors $b_1^d = (1/\sqrt{2}, 0, 0)$ and $(-1/\sqrt{2}, 0, 0)$, representing heading angles of $\pm 45^\circ$ while upright, with a linear transition between the two in a step-like function (shown in Fig. 5). We define the heading angle θ as the angle between the heading vector b_1 (aligned with the x -axis in Fig. 1) and the global y -axis, such that $\theta := \arctan \frac{b_1(2)}{b_1(1)}$, with desired heading θ_d . The vehicle stabilized at θ_d set-points, maintaining hovering stability and converging to θ_d with a maximum heading error of 14.2° after convergence. Additionally, during the transitions in θ_d between set-points (shown in gray), the vehicle tracked the π rad/s angular velocity of θ_d . In Fig. 5, we also report forces and torques produced by the vehicle during its response (estimated according to Section IV-A). We observe that the RoboBee exhibits controllable yaw torque in flight; the first such demonstration for this vehicle.

D. Lateral Control With Active Heading Tracking

We now demonstrate that the RoboBee can achieve full control authority (controllable thrust and body torques, with attitude and lateral control) using these methods. We validate full control authority of the vehicle with a demonstration of a maneuver wherein the RoboBee circles (laterally) about a center point, with its gaze fixed on that point; $r_d := (\cos(2\pi/5t) + x_d, \sin(2\pi/5t) + y_d, z_d)$ which creates a circle with a period of five seconds, centered at $c := (x_d, y_d, z_d)$, and the heading vector is defined as $b_1^d := c - r_d$. Results of this maneuver are shown in Fig. 6. We observe a position error (reported as mean $\pm$ standard deviation) of -0.68 ± 2.40 cm in the x direction, 1.78 ± 2.23 cm in the y direction, and 0.167 ± 0.561 cm in the z direction (corresponding to an effective center-point shift of 2.04 cm, with a circle radius expanded by 2.96 cm.) This performance is comparable to the previous state-of-the-art for RoboBee control [5], which is notable given that the applied controller was not designed

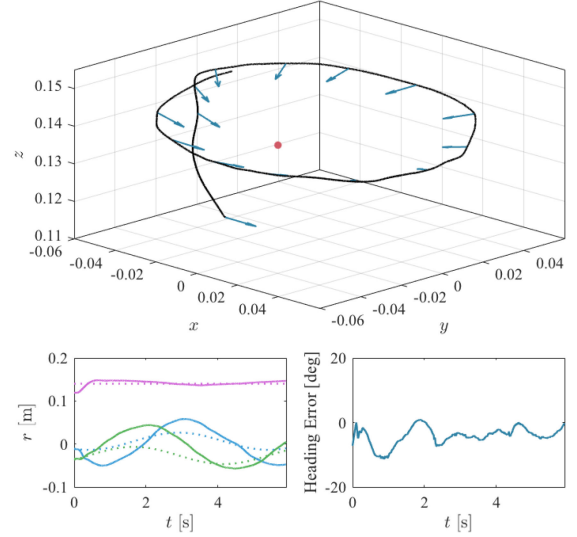


Fig. 6. Controlled flight of the RoboBee circling and facing a center point in midair; (top) Three-dimensional vehicle trajectory in black, with the observed heading in blue and the desired center point in red; (bottom left) Desired (dotted) and observed position, with x in blue, y in green, and z in magenta; and (bottom right) heading angle error.

for FWMAVs, nor does it include adaptive elements. Additionally, [5] does not achieve active heading control; in this demonstration, the vehicle tracks its desired heading with an error of $-4.13 \pm 2.70^\circ$.

VI. CONCLUSION

FWMAV modeling may complement control strategies both by capturing vehicle asymmetries and giving an analytic, physically intuitive understanding of its dynamics. We present a FWMAV dynamics model which allows analytical wingstroke and vehicle dynamics descriptions and closed-form nonlinear expressions describing force and torque generation via quasi-steady analysis, while respecting inherent vehicle asymmetries. We validate our model with experimental data used to characterize the vehicle, and demonstrate the utility of this system identification process in hovering and heading tracking via a geometric controller. In applying these methods, we demonstrate in-flight heading control of the RoboBee for the first time, and achieve full control authority with simultaneous attitude and lateral control. These modeling, system identification, and control processes provide marked advantages in control authority for this vehicle, and may be applied on a wider scale to other FWMAV designs. We anticipate that with the inclusion of an adaptive law, the controller in conjunction with our model and system identification process would further improve the tracking performance.

ACKNOWLEDGMENT

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

REFERENCES

- [1] J. P. Whitney and R. J. Wood, "Aeromechanics of passive rotation in flapping flight," *J. Fluid Mech.*, vol. 660, pp. 197–220, 2010.
- [2] K. Y. Ma, S. M. Felton, and R. J. Wood, "Design, fabrication, and modeling of the split actuator microrobotic bee," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2012, pp. 1133–1140.
- [3] Z. Tu, F. Fei, J. Zhang, and X. Deng, "An at-scale tailless flapping-wing hummingbird robot. I. Design, optimization, and experimental validation," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1511–1525, Oct. 2020.
- [4] M. Graule *et al.*, "Perching and takeoff of a robotic insect on overhangs using switchable electrostatic adhesion," *Science*, vol. 352, no. 6288, pp. 978–982, 2016.
- [5] P. Chirarattananon, K. Y. Ma, and R. J. Wood, "Adaptive control of a millimeter-scale flapping-wing robot," *Bioinspiration Biomimetics*, vol. 9, no. 2, 2014, Art. no. 025004.
- [6] Y. M. Chukewad and S. Fuller, "Yaw control of a hovering flapping-wing aerial vehicle with a passive wing hinge," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 1864–1871, Apr. 2021.
- [7] J. Zhang, Z. Tu, F. Fei, and X. Deng, "Geometric flight control of a hovering robotic hummingbird," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2017, pp. 5415–5421.
- [8] X. Yang, Y. Chen, L. Chang, A. A. Calderón, and N. O. Pérez-Arancibia, "Bee : A 95-mg four-winged insect-scale flying robot driven by twinned unimorph actuators," *IEEE Robot. Automat. Lett.*, vol. 4, no. 4, pp. 4270–4277, Oct. 2019.
- [9] N. Gravish and R. J. Wood, "Anomalous yaw torque generation from passively pitching wings," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 3282–3287.
- [10] Z. Tu, F. Fei, L. Liu, Y. Zhou, and X. Deng, "Flying with damaged wings: The effect on flight capacity and bio-inspired coping strategies of a flapping wing robot," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 2114–2121, Apr. 2021.
- [11] D. B. Doman, M. W. Oppenheimer, and D. O. Sigthorsson, "Wing-beat shape modulation for flapping-wing micro-air-vehicle control during hover," *J. Guidance, Control, Dyn.*, vol. 33, no. 3, pp. 724–739, 2010.
- [12] P. Chirarattananon, K. Y. Ma, and R. J. Wood, "Perching with a robotic insect using adaptive tracking control and iterative learning control," *Int. J. Robot. Res.*, vol. 35, no. 10, pp. 1185–1206, 2016.
- [13] N. P. Hyun, R. McGill, R. J. Wood, and S. Kuindersma, "A new control framework for flapping-wing vehicles based on 3D pendulum dynamics," *Automatica*, vol. 123, 2021, Art. no. 109293.
- [14] R. Steinmeyer, P. H. Nak-seung, E. F. Helbling, and R. J. Wood, "Yaw torque authority for a flapping-wing micro-aerial vehicle," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 2481–2487.
- [15] J. P. Boyd, "Computing the zeros, maxima and inflection points of chebyshev, legendre and fourier series: Solving transcendental equations by spectral interpolation and polynomial rootfinding," *J. Eng. Math.*, vol. 56, no. 3, pp. 203–219, 2006.
- [16] X. Deng, L. Schenato, W. C. Wu, and S. S. Sastry, "Flapping flight for biomimetic robotic insects: Part I-system modeling," *IEEE Trans. Robot.*, vol. 22, no. 4, pp. 776–788, Aug. 2006.
- [17] P. Chirarattananon, K. Y. Ma, and R. J. Wood, "Single-loop control and trajectory following of a flapping-wing microrobot," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 37–44.
- [18] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE (3)," in *Proc. IEEE Conf. Decis. Control*, 2010, pp. 5420–5425.
- [19] C. T. Orłowski and A. R. Girard, "Modeling and simulation of nonlinear dynamics of flapping wing micro air vehicles," *AIAA J.*, vol. 49, no. 5, pp. 969–981, 2011.
- [20] L. Chang, A. Rigo, and N. O. Pérez-Arancibia, "Automatic multigrid generation for an unstructured parallel overset-grid solver," in *Proc. AIAA Scitech Forum*, 2020, Art. no. 1406.
- [21] M. H. Dickinson, F.-O. Lehmann, and S. P. Sane, "Wing rotation and the aerodynamic basis of insect flight," *Science*, vol. 284, no. 5422, pp. 1954–1960, 1999.
- [22] B. M. Finio, N. O. Pérez-Arancibia, and R. J. Wood, "System identification and linear time-invariant modeling of an insect-sized flapping-wing micro air vehicle," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2011, pp. 1107–1114.
- [23] N. T. Jafferis, E. F. Helbling, M. Karpelson, and R. J. Wood, "Untethered flight of an insect-sized flapping-wing microscale aerial vehicle," *Nature*, vol. 570, no. 7762, pp. 491–495, 2019.